

Project Title
Machine Learning-Based Alzheimer's Disease
Classification Using MRI and Clinical Data

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Technologies: Python, Scikit-learn, XGBoost, Pandas
GitHub Repository

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Abstract

Alzheimer's disease (AD) is a progressive neurodegenerative disorder that requires timely and accurate diagnosis to improve patient management and treatment planning. This project investigates the application of machine learning techniques for the multi-class classification of Alzheimer's disease stages using non-invasive MRI measurements, cognitive assessment scores, demographic information, and clinical data. A comprehensive machine learning pipeline was developed that included data preprocessing, missing value imputation, feature encoding, feature scaling, feature selection, and model evaluation.

Two ensemble learning algorithms, Random Forest and Extreme Gradient Boosting (XGBoost), were implemented and compared to assess their effectiveness in predicting Alzheimer's disease stages. Model performance was evaluated using accuracy, precision, recall, F1-score, confusion matrices, Receiver Operating Characteristic (ROC) curves, Precision-Recall curves, and cross-validation. Experimental results showed that XGBoost achieved slightly better overall classification performance than Random Forest, demonstrating its ability to capture complex relationships within heterogeneous clinical data. The study also identified challenges in accurately classifying minority disease stages, highlighting the impact of class imbalance on predictive performance.

The findings demonstrate the potential of machine learning to support non-invasive clinical decision-making for Alzheimer's disease diagnosis and staging. Future work may focus on incorporating deep learning techniques, multimodal medical imaging, advanced feature engineering, and larger, more balanced datasets to further improve model robustness and predictive accuracy.

1. Introduction

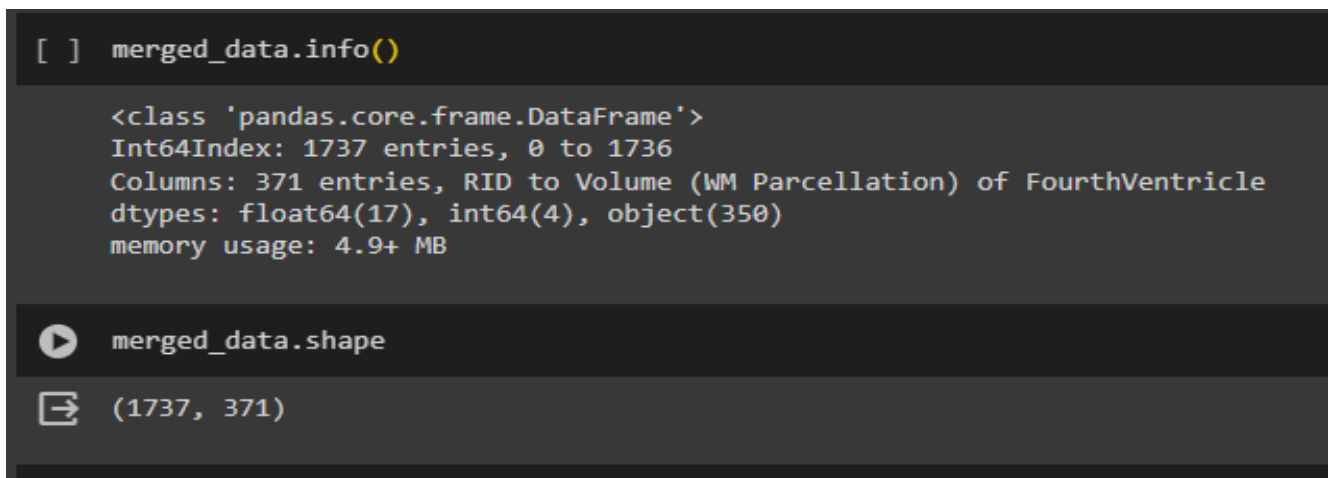
Alzheimer's disease (AD) is the most common neurodegenerative disorder and the 6th leading cause of death in the United States. The world-wide disease burden of AD is projected to reach \$2 trillion by 203015, which necessitates early detection. Despite extensive research and advances in clinical practice, less than 50% of the AD patients are diagnosed accurately for their pathology and disease progression based on their clinical symptoms. The most conclusive evidence for AD are the presence of amyloid plaques and neurofibrillary tangles in histopathology.

The assessment of amyloid plaques is performed by an amyloid-PET examination, in which a radiotracer that binds to the amyloid protein is injected intravenously in a patient, and the position and quantity is evaluated.

However, the procedure is expensive and invasive, so it is a matter of ongoing research the possibility of diagnosing and grading AD without using PET examination, but rather relying on brain MRI data, cognitive scores, demographic characteristics and genetic risk.

2. Data Description

Alzheimer's disease (AD) is a major neurodegenerative disorder with significant morbidity and mortality. Given its projected global impact and the limitations of current diagnostic practices, there is a strong emphasis on improving early detection methods. The gold standard for AD diagnosis involves the assessment of amyloid plaques via amyloid-PET scans. However, due to the costs and invasiveness of PET scans, there is ongoing research into alternative diagnostic approaches that leverage non-invasive methods like MRI, along with cognitive assessments, demographics, and genetic information. A subset of the Alzheimer's Disease Neuroimaging Initiative (ADNI) database (adni.loni.usc.edu) data preprocessed for the TADPOLE Grand Challenge is used for the work.



```
[ ] merged_data.info()

<class 'pandas.core.frame.DataFrame'>
Int64Index: 1737 entries, 0 to 1736
Columns: 371 entries, RID to Volume (WM Parcellation) of FourthVentricle
dtypes: float64(17), int64(4), object(350)
memory usage: 4.9+ MB

merged_data.shape

(1737, 371)
```

Figure 1.1: Data description, attributes and data types.

The available ADNI data contains baseline imaging, clinical, and genetic data for over 1737 patients originating from two studies (ADNI1, ADNI2).

2.1 Value Ranges

```
[47] merged_data.describe(include='all')
```

	RID	Test_data	Diagnosis	FDG_PET	CDRSB	ADAS11	ADAS13	MMSE	RAVLT_immediate	RAVLT_learning	...	Cortical Thickness Standard Deviation of Rightlingual
count	1737.000000	1737.000000	1737	1303.000000	1737.000000	1732.000000	1723.000000	1737.000000	1731.000000	1731.000000	...	1737.000
unique	NaN	NaN	5	NaN	NaN	NaN	NaN	NaN	NaN	NaN	...	307.000
top	NaN	NaN	LMCI	NaN	NaN	NaN	NaN	NaN	NaN	NaN	...	0.588
freq	NaN	NaN	562	NaN	NaN	NaN	NaN	NaN	NaN	NaN	...	19.000
mean	2603.533679	0.199770	NaN	1.233933	1.639321	10.773672	16.899008	27.173287	35.153091	4.177354	...	NaN
std	1926.356831	0.399942	NaN	0.154161	1.792005	6.657689	9.584230	2.666661	12.464480	2.734129	...	NaN
min	2.000000	0.000000	NaN	0.697264	0.000000	0.000000	0.000000	18.000000	0.000000	-4.000000	...	NaN
25%	759.000000	0.000000	NaN	1.141735	0.000000	6.000000	9.330000	26.000000	26.000000	2.000000	...	NaN
50%	2150.000000	0.000000	NaN	1.245480	1.000000	9.000000	15.000000	28.000000	34.000000	4.000000	...	NaN
75%	4546.000000	0.000000	NaN	1.336715	2.500000	14.000000	23.000000	29.000000	44.000000	6.000000	...	NaN
max	5296.000000	1.000000	NaN	1.707170	10.000000	42.670000	54.670000	30.000000	71.000000	12.000000	...	NaN

11 rows x 371 columns

Figure 2. Value ranges of Numeric Attributes

```
7] merged_data.describe(include='all')
```

Volume (Cortical Parcellation) of RightMedialOrbitofrontal	Surface Area of RightMedialOrbitofrontal	Cortical Thickness Average of RightMedialOrbitofrontal	Cortical Thickness Standard Deviation of RightMedialOrbitofrontal	Volume (Cortical Parcellation) of RightMiddleTemporal	Surface Area of RightMiddleTemporal
1737	1737	1737	1737.00	1737	1737
1279	769	727	426.00	1498	1046
			0.71		
16	16	16	16.00	16	16
NaN	NaN	NaN	NaN	NaN	NaN
NaN	NaN	NaN	NaN	NaN	NaN
NaN	NaN	NaN	NaN	NaN	NaN
NaN	NaN	NaN	NaN	NaN	NaN
NaN	NaN	NaN	NaN	NaN	NaN
NaN	NaN	NaN	NaN	NaN	NaN
NaN	NaN	NaN	NaN	NaN	NaN

Figure 3. Value ranges of Numeric Attributes

Cortical Thickness Average of RightMedialOrbitofrontal	Cortical Thickness Standard Deviation of RightMedialOrbitofrontal	Volume (Cortical Parcellation) of RightMiddleTemporal	Surface Area of RightMiddleTemporal	Cortical Thickness Average of RightMiddleTemporal	Cortical Thickness Standard Deviation of RightMiddleTemporal	Volume (WM Parcellation) of FourthVentricle
1737	1737.00	1737	1737	1737	1737.000	1737
727	426.00	1498	1046	744	344.000	1188
	0.71				0.676	
16	16.00	16	16	16	17.000	16
NaN	NaN	NaN	NaN	NaN	NaN	NaN
NaN	NaN	NaN	NaN	NaN	NaN	NaN
NaN	NaN	NaN	NaN	NaN	NaN	NaN
NaN	NaN	NaN	NaN	NaN	NaN	NaN
NaN	NaN	NaN	NaN	NaN	NaN	NaN
NaN	NaN	NaN	NaN	NaN	NaN	NaN
NaN	NaN	NaN	NaN	NaN	NaN	NaN

Figure 4. Value ranges of Numeric Attributes

The figures above (2,3,4) show the value ranges of all the attributes in numeric form.

The dataset's potential role in researching alternative diagnostic methodologies for Alzheimer's disease, emphasizing the need for and interest in less invasive, more accessible diagnostic tools. This dataset could be invaluable for developing predictive models or for understanding the correlations between non-invasive measurements and AD pathology.

This sheet outlines abbreviations for each class along with their full descriptions:

Sr. No.	Class Abbreviation	Description
1	CN	Cognitive normal
2	SMC	Significant Memory Concern
3	EMCI	Early mild cognitive impairment
4	MCI	Mild cognitive impairment
5	LMCI	Late mild cognitive impairment

Table 1. Abbreviations of all five target classes

3. Literature Review

Applying Convolutional Neural Networks for Pre-detection of Alzheimer's Disease from Structural MRI Data (2017): This study examines the use of CNNs for the early detection of AD from MRI data, suggesting that deep learning can significantly enhance diagnostic accuracy in the pre-detection phase of Alzheimer's, with a model achieving around 96% accuracy.

A Deep Learning Model for Early Diagnosis of Alzheimer's Disease and Dementia From MR Images (2021): This paper focuses on a deep learning approach for early AD diagnosis, using advanced neural network architectures to analyze MR images. The results highlight the potential of deep learning to improve diagnostic processes and outcomes in AD.

Alzheimer's Disease Classification Using Pre-trained Deep Networks (2021): This research explores the application of pre-trained deep networks for Alzheimer's disease classification, demonstrating how leveraging existing deep learning models can facilitate accurate AD diagnosis, even with limited domain-specific training data.

A Transfer Learning Approach for Early Diagnosis of Alzheimer's (2021): Investigating the efficacy of transfer learning for AD diagnosis, this paper shows that applying knowledge from pre-trained models to AD-specific datasets can significantly improve diagnostic accuracy, emphasizing the value of transfer learning in medical imaging analysis.

Brain MRI Analysis for Alzheimer's Disease Diagnosis Using an Ensemble System of Deep Convolutional Neural Networks (2018): This study introduces an ensemble of CNNs to analyze brain MRI for AD diagnosis, indicating that combining multiple deep learning models can enhance predictive performance, providing a robust approach to identifying Alzheimer's from MRI scans.

Prediction of Alzheimer's Disease from MRI Images Using Deep Learning (2020): The research presents a deep learning framework to predict AD from MRI images, attesting to the high accuracy of models like 2D CNN and VGG16 in diagnosing Alzheimer's, underscoring deep learning's potential in medical image analysis.

Analysis of MRI Image Data (2023): This most recent paper delves into sophisticated MRI image analysis using deep learning to diagnose Alzheimer's disease, showcasing cutting-edge methods in extracting and interpreting complex patterns in MRI data for more accurate and early detection of AD.

3.1 key findings

The research papers show that using deep learning and machine learning on MRI scans can help diagnose Alzheimer's disease better. They found that techniques like convolutional neural networks (CNNs) and transfer learning can spot changes in the brain linked to Alzheimer's with high accuracy and can find it early. Using pre-trained networks and transfer learning makes this process more efficient. Also, combining different types of data or methods improves diagnosis, and catching Alzheimer's early is crucial for effective treatment. Advanced techniques for analyzing MRI images mentioned in the papers show ongoing progress in understanding Alzheimer's. Overall, these studies indicate that AI tools can become more accurate and efficient in diagnosing Alzheimer's early by using complex deep learning methods on MRI scans.

3.2 Recommendations

- Diverse Datasets: Use varied datasets with different patient characteristics and MRI types to improve model accuracy across different groups.
- Transparent Models: Make models easier to understand, so clinicians trust them more and can integrate them better into practice.
- Multimodal Data Integration: Combine MRI data with other types like genetics and cognition for a more complete Alzheimer's diagnostic tool.
- Cross-disciplinary Collaboration: Work together across fields like neurology, radiology, and data science to create models that fit real-world needs and knowledge.
- Rigorous Validation: Test models thoroughly in clinical settings to ensure they work well for different patients and settings.
- Continuous Learning: Build models that can keep learning and improving as more data and knowledge about Alzheimer's become available.

4. Research Question

Given the dataset and its focus on Alzheimer's disease diagnosis using machine learning, we can define several research questions that address critical challenges in the detection and staging of Alzheimer's disease. Here are potential research questions, each directly linked to the provided dataset and the ultimate goal of improving diagnosis accuracy and patient care:

1. Can machine learning models predict Alzheimer's disease stages using non-invasive MRI and cognitive test data with a high degree of accuracy?

Dataset Reference: Utilizes MRI data and cognitive scores from the "Cognitive score" and "Data" sheets.

Expected Outcome: Develop a predictive model that classifies patients into one of the six Alzheimer's disease stages or three broader categories with high accuracy, precision, and recall metrics. The outcome will help in early and accurate diagnosis of Alzheimer's disease, potentially leading to better patient management and care.

2. How effective are demographic and genetic risk factors in improving the predictive accuracy of Alzheimer's disease progression models?

Dataset Reference: Incorporates demographic information (Age, Gender, Education) and genetic data (high-risk alleles) from the "Data" sheet alongside MRI and cognitive data.

Expected Outcome: Analyze the impact of including demographic and genetic information in the predictive models. The goal is to determine if these additional data points significantly enhance model performance in terms of specificity and sensitivity.

3. Can longitudinal changes in cognitive scores predict the transition between different Alzheimer's disease stages?

Dataset Reference: Focus on changes in cognitive scores over time, assuming longitudinal data is available or can be simulated from the "Cognitive score" sheet.

Expected Outcome:

Identify patterns in cognitive score changes linked to disease progression, potentially creating a model to predict how Alzheimer's advances, helping clinicians anticipate and manage changes in patients.

Use machine learning to fill gaps in Alzheimer's diagnosis and monitoring. The aim is to improve clinical practices with better, non-invasive methods for diagnosing and tracking the disease, improving patient care and outcomes.

We are going to address question 1 only.

5. Data Analysis

Here is the statistical analysis shown in figure 5, of the feature set. This data set consists of 371 features. So there is just a snippet of a few features but details can be viewed in the code file provided with this report.

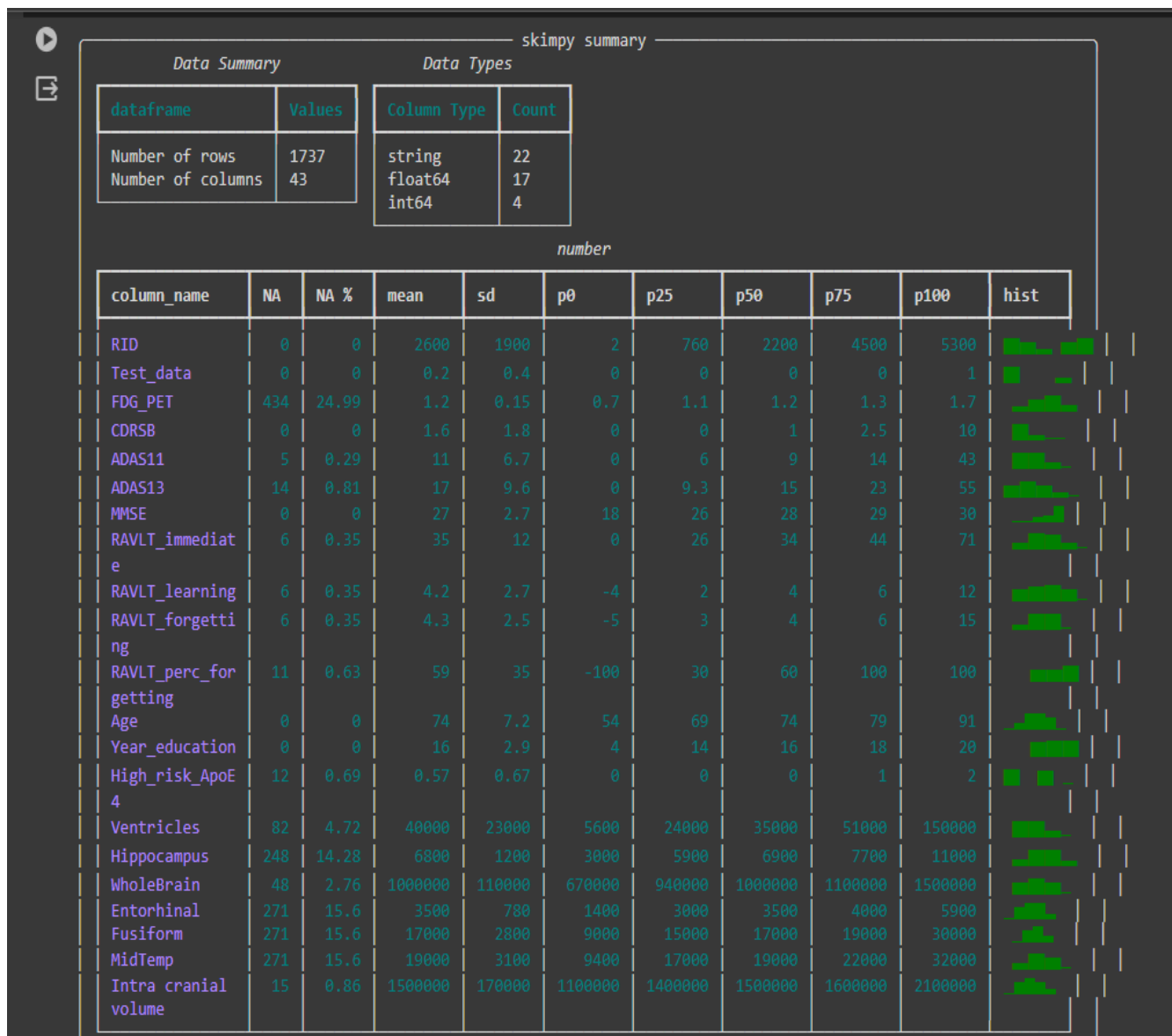


Figure 5. Statistical data description of numerical attributes

string				
column_name	NA	NA %	words per row	total words
Diagnosis	0	0	1	1737
Gender	0	0	1	1737
Ethnicity	0	0	2	3405
Race	0	0	1	1778
Marital_status	0	0	1	1801
Volume (WM Parcellation) of RightUndetermined	0	0	2	3474
Volume (WM Parcellation) of RightWMHypoIntensities	0	0	2	3474
Volume (Cortical Parcellation) of LeftCorpusCallosum	0	0	2	3474
Surface Area of LeftCorpusCallosum	0	0	2	3474
Cortical Thickness Average of LeftCorpusCallosum	0	0	2	3474
Cortical Thickness Standard Deviation of LeftCorpusCallosum	0	0	2	3474
Volume (Cortical Parcellation) of LeftHemisphereWM	0	0	2	3474
Volume (WM Parcellation) of LeftInterior	0	0	2	3474
Volume (WM Parcellation) of LeftNonWMHypoIntensities	0	0	2	3474
Volume (WM Parcellation) of LeftUndetermined	0	0	2	3474
Volume (WM Parcellation) of LeftWMHypoIntensities	0	0	2	3474
Volume (Cortical Parcellation) of RightCorpusCallosum	0	0	2	3474
Surface Area of RightCorpusCallosum	0	0	2	3474
Cortical Thickness Average of RightCorpusCallosum	0	0	2	3474
Cortical Thickness Standard Deviation of RightCorpusCallosum	0	0	2	3474
Volume (Cortical Parcellation) of RightHemisphereWM	0	0	2	3474
Volume (WM Parcellation) of RightInterior	0	0	2	3474

Figure 6. Description of Non-numeric attributes

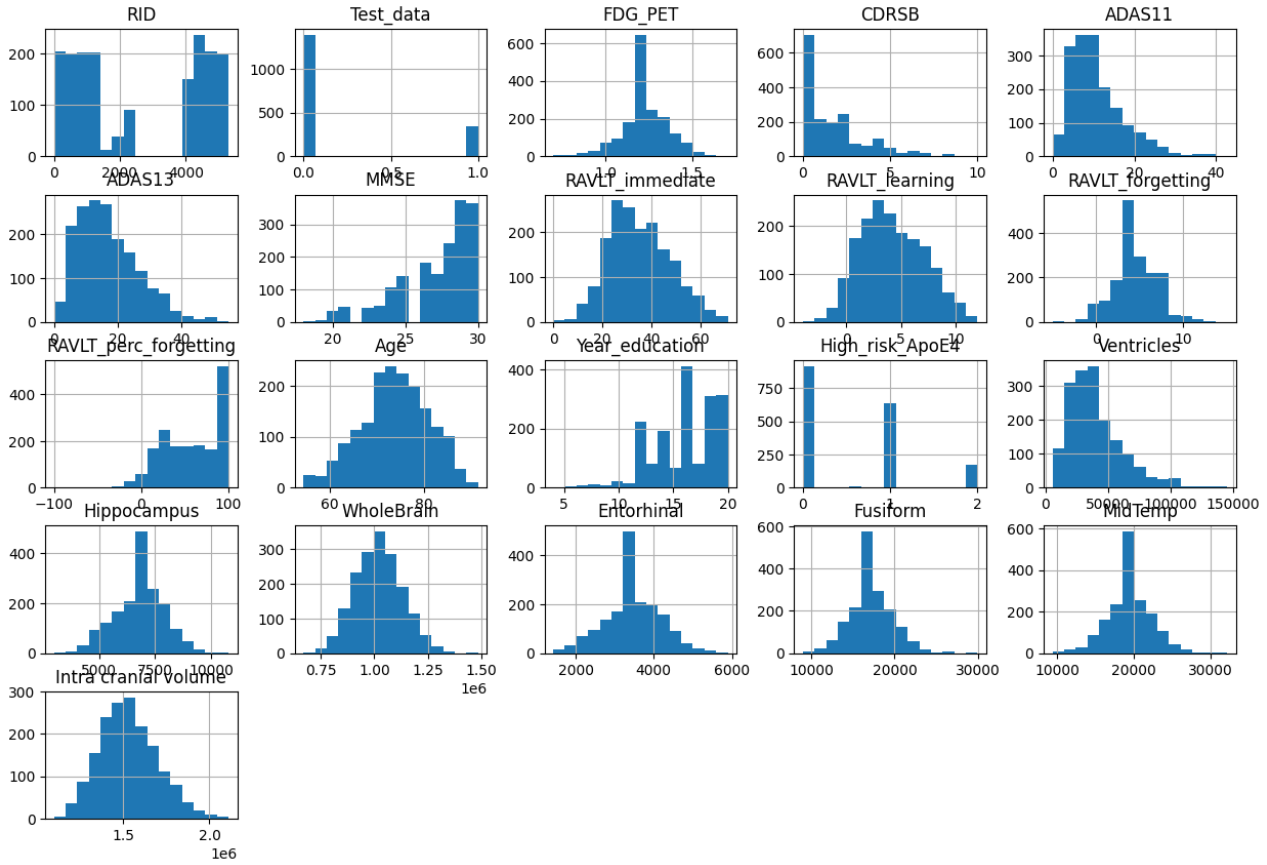


Figure 7. The histogram of numeric features

The figure 7 represents histograms indicating skewed distributions in key volumetric and cognitive features, which could necessitate data normalization or transformation when preparing inputs for predictive modeling to meet algorithmic assumptions and improve model reliability.

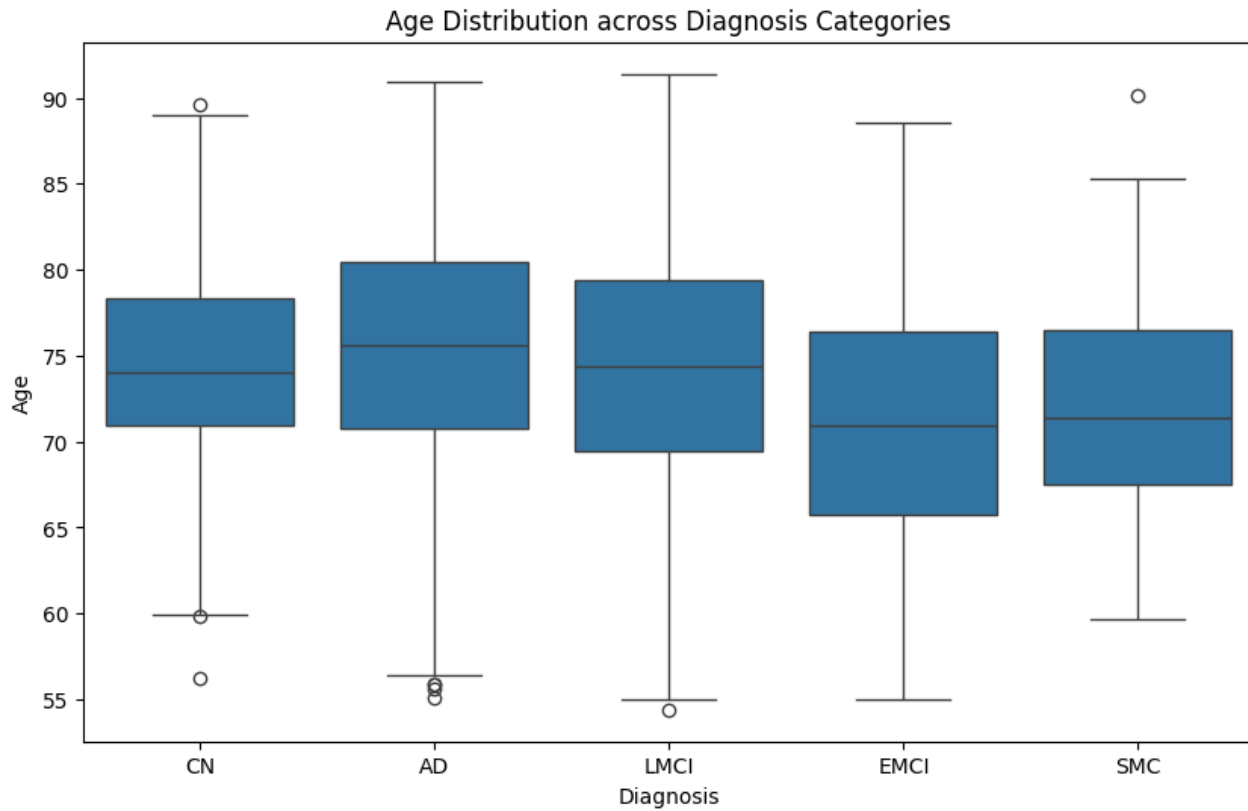


Figure 8. Box plot showing the Alzheimer disease for different groups of ages for six stages.

The figure 8 illustrates the box plot dividing different age groups with respect to the 6 categories of the disease. The age distribution across diagnosis categories shows a consistency in median ages but with some outliers, particularly in the cognitively normal and Alzheimer's groups. This could influence model accuracy and requires careful statistical management.

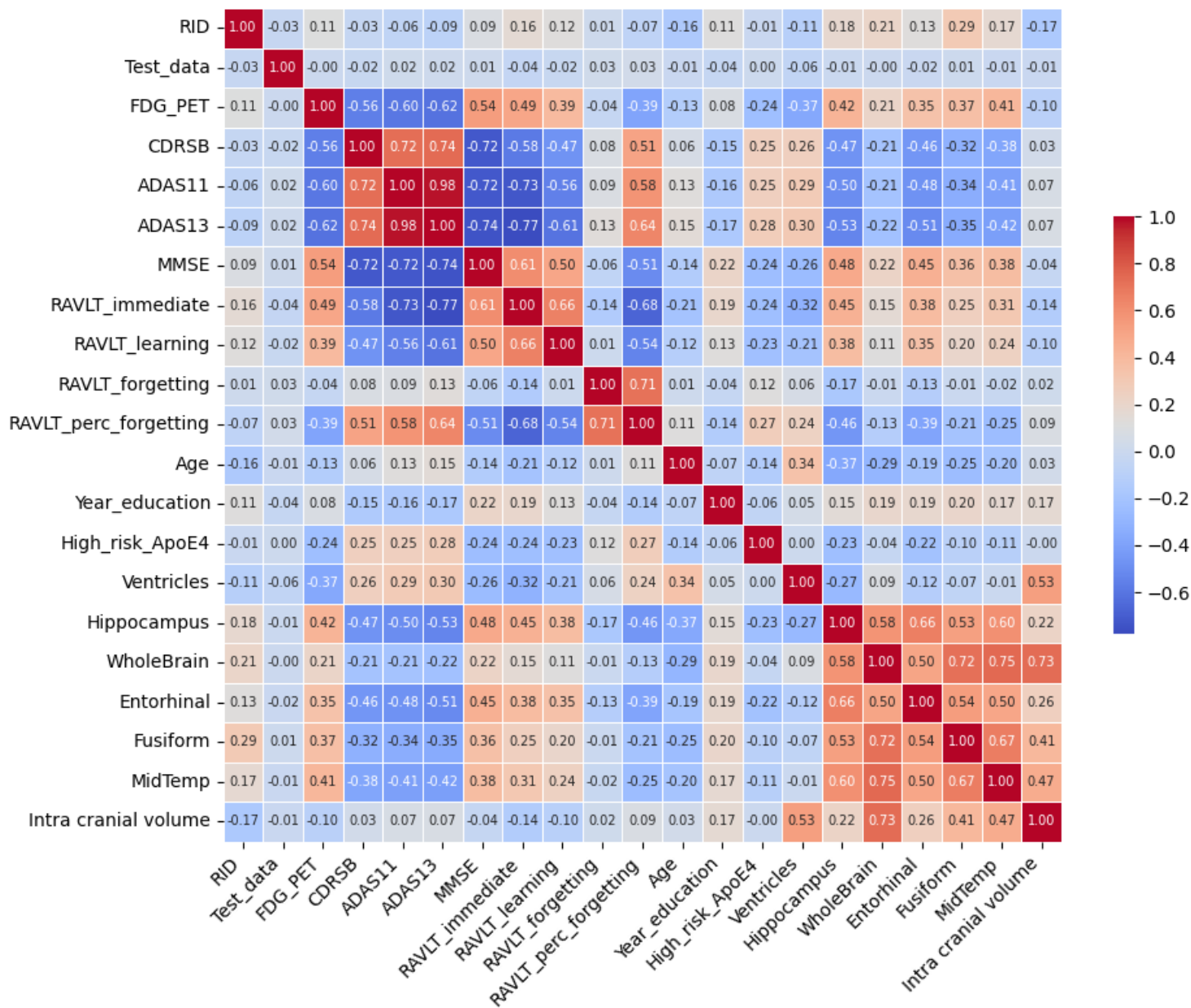


Figure 9. The correlation matrix, depicting the association of features.

Correlation of each class

The correlation matrix suggests a strong association between cognitive test scores and volumetric measurements like hippocampal size, indicating their importance in disease assessment. There's also a noticeable inverse relationship between cognitive scores and ventricle size, signifying disease progression.

Key actions from this analysis include prioritizing relevant features for model inclusion, addressing data skewness and outliers and considering multimodal approaches that combine structural MRI metrics with

cognitive scores to enhance predictive accuracy. These steps are fundamental in constructing robust models aimed at early and accurate diagnosis of Alzheimer's disease.

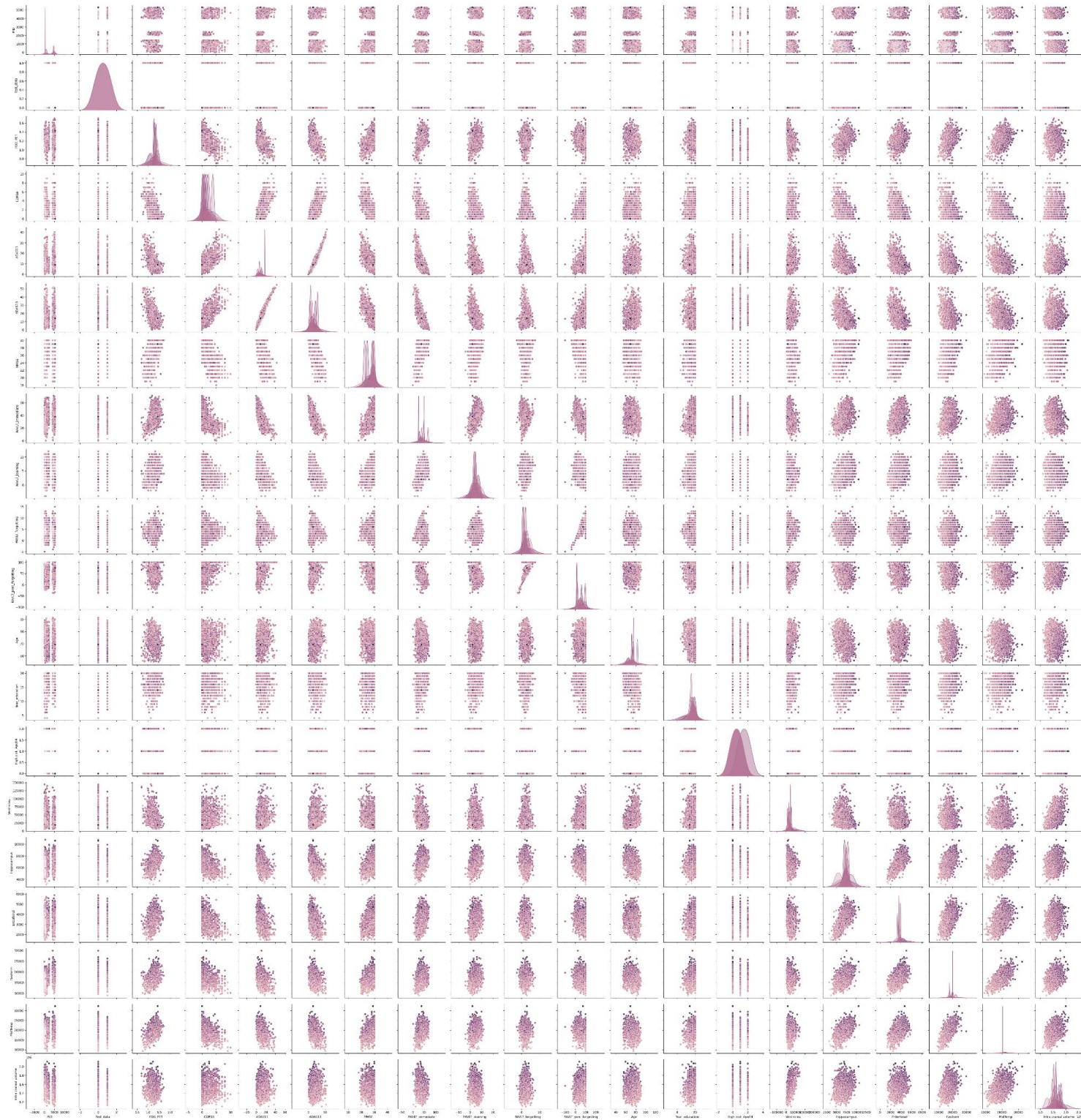


Figure 10. The graph shows the correlation among various features, Hippocampus, Entorhinal, fusiform, MidTemp, IntraCranial volume, FDG_PET is related with Fusiform, Ventricle has positive correlation with Intracranial volume. RAVLT Pulse forgetting and RAVLT forgetting are also correlated.

The figure 10, scatter plot of numeric feature set illustrates the positive and negative correlation among many features for example Hippocampus, Entorhinal, fusiform, MidTemp, IntraCranial volume have positive correlation with MidTemp, FDG_PET is positively related with Fusiform, Ventricles has positive correlation with Intracranial volume. RAVLT Pulse forgetting and RAVL forgetting are also correlated.

6. Choice of Solution techniques and Model

For the challenge of predicting Alzheimer's disease stages using machine learning models with MRI and cognitive test data, the choices of algorithms, tools, and processes can be justified based on several factors, including the nature of the data, the requirements of the problem, and insights from the literature. Here's a detailed breakdown:

6.1 Choice of Machine Learning Algorithms

Random Forest and Gradient Boosting Machines (XGBoost):

- **Rationale:** Both Random Forest and XGBoost are robust to overfitting and can handle various types of data effectively. They are particularly suitable for complex classification tasks like predicting Alzheimer's stages because they can model nonlinear relationships and interactions between features.
- **Literature Support:** These algorithms have shown promising results in similar healthcare data settings. For instance, ensemble methods have been effective in classifying stages of diseases based on heterogeneous data sources like imaging and clinical data.

6.2 Choice of Data Preprocessing Techniques

a) Imputation :

- **Rationale:** Missing data is common in clinical datasets. Using sophisticated imputation strategies like IterativeImputer helps maintain data integrity by estimating missing values based on other features, which is crucial for accurate predictions.
- **Literature Support:** Iterative imputation has been recommended for complex datasets where each attribute can help estimate others, common in medical datasets.

b) Normalization and Encoding:

- **Rationale:** Standardizing features (using StandardScaler) ensures that no single feature dominates due to its scale, which is important for models like SVM and neural networks. Encoding categorical data (using OneHotEncoder) transforms it into a format that can be effectively used by machine learning algorithms.
- **Literature Support:** Normalization and encoding are standard practices in data preprocessing to prepare datasets for effective machine learning modeling, ensuring comparability of results and effectiveness of algorithms.

6.3 Choice of Tools

a) Scikit-learn and XGBoost Libraries:

- **Rationale:** These libraries offer robust, efficient implementations of a wide range of machine learning algorithms along with extensive support for preprocessing, model tuning, and evaluation. Their widespread use and active community support make them suitable for both prototyping and production.
- **Literature Support:** These tools are widely used in academic and industrial research, noted for their reliability and effectiveness in various studies across health informatics.

6.4 Choice of Model Evaluation and Validation Processes

a) Cross-Validation:

- **Rationale:** Using 5-fold cross-validation ensures that the model's performance is tested on different subsets of the dataset, providing a more generalized performance estimate.
- **Literature Support:** Cross-validation is a standard practice in machine learning for healthcare applications to ensure that models do not overfit and can generalize well to new, unseen data.

b) Performance Metrics (Accuracy, Precision, Recall, F1-score, ROC AUC):

- **Rationale:** These metrics provide a comprehensive view of model performance, especially in a medical classification context where the balance between sensitivity (recall) and specificity (precision) is crucial.
- **Literature Support:** Commonly used in medical diagnosis challenges, these metrics help in understanding the trade-offs in different types of errors, which is critical for clinical applications.

7. Practical Work

To answer the research question, *"Can machine learning models predict Alzheimer's disease stages using non-invasive MRI and cognitive test data with a high degree of accuracy?"* We have developed a comprehensive machine learning pipeline using Python and libraries such as scikit-learn and

XGBoost. Below, I is the outline of steps involved, from data preprocessing to model evaluation, ensuring that each phase is thoroughly documented:

7.1. Data Preprocessing

Loading and Initial Cleaning:

- Data is initially loaded from an Excel file containing MRI data and cognitive scores.
- Irrelevant or redundant features (e.g., identifiers not related to the outcome) are removed.
- Data integrity checks are performed to identify missing values or corrupted entries.

Handling Missing Values:

- IterativeImputer from scikit-learn is used for numerical features to estimate missing values based on other features in the dataset, employing a RandomForestRegressor as the estimator. This method is chosen due to its ability to handle complex patterns and interactions in data, which is often seen in medical datasets.
- Categorical features with missing values are imputed using the most frequent category via SimpleImputer.

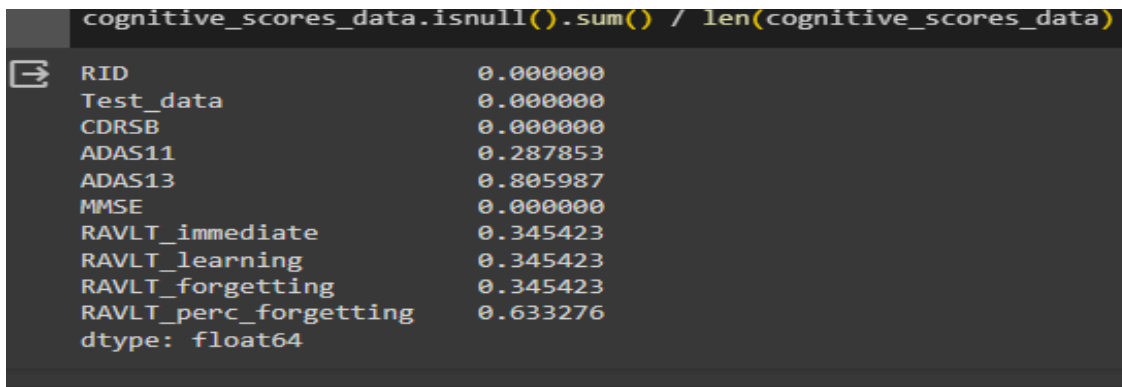


Figure 7.1 : Missing Values in percentage

Feature Scaling and Encoding:

- Numerical features are standardized using StandardScaler to ensure all features contribute equally to model predictions, critical for distance-based algorithms.
- Categorical features are encoded using OneHotEncoder, which transforms categorical variables into a format that can be provided to ML algorithms to better process.

Feature Selection:

- An initial model is used to assess feature importance. Features with very low importance (below a threshold, e.g., 0.01) are removed from further analysis to reduce model complexity and overfitting.

7.2. Model Training and Validation

Splitting Data:

- The dataset is split into training and testing sets, with 80% of data used for training and 20% reserved for testing. The split is stratified based on the Test_data column to ensure that the distribution of disease stages is consistent across both datasets.

Model Selection and Training:

- A RandomForestClassifier is selected due to its robustness and effectiveness in handling tabular data like MRI and cognitive scores. It is also less prone to overfitting compared to more complex models.
- The model is trained using the training set within a pipeline that includes both preprocessing steps and the classifier itself.

Hyperparameter Tuning:

- GridSearchCV is utilized to find the optimal parameters for the RandomForest model, including n_estimators, max_depth, min_samples_split, and min_samples_leaf. This step is crucial to enhance model performance by exploring a range of parameter values.

7.3. Model Evaluation

Performance Metrics:

- The model's performance is assessed using accuracy, precision, recall, and F1-score. These metrics are chosen to provide a balanced view of model performance across all classes of Alzheimer's disease stages.
- Confusion Matrix: Provides insight into the classification accuracy across different stages, helping to identify if any particular stage is consistently misclassified.

Cross-Validation:

- To ensure the model's robustness and generalizability, k-fold cross-validation is conducted during the grid search. This method provides a more reliable estimate of the model's performance on unseen data.

ROC and Precision-Recall Curves:

- For a more detailed evaluation, ROC curves for each class are plotted to assess the trade-offs between true positive rate and false positive rate.
- Precision-recall curves are also generated, especially important in scenarios where classes are imbalanced.

8. Results

8.1 XGBoost Performance:

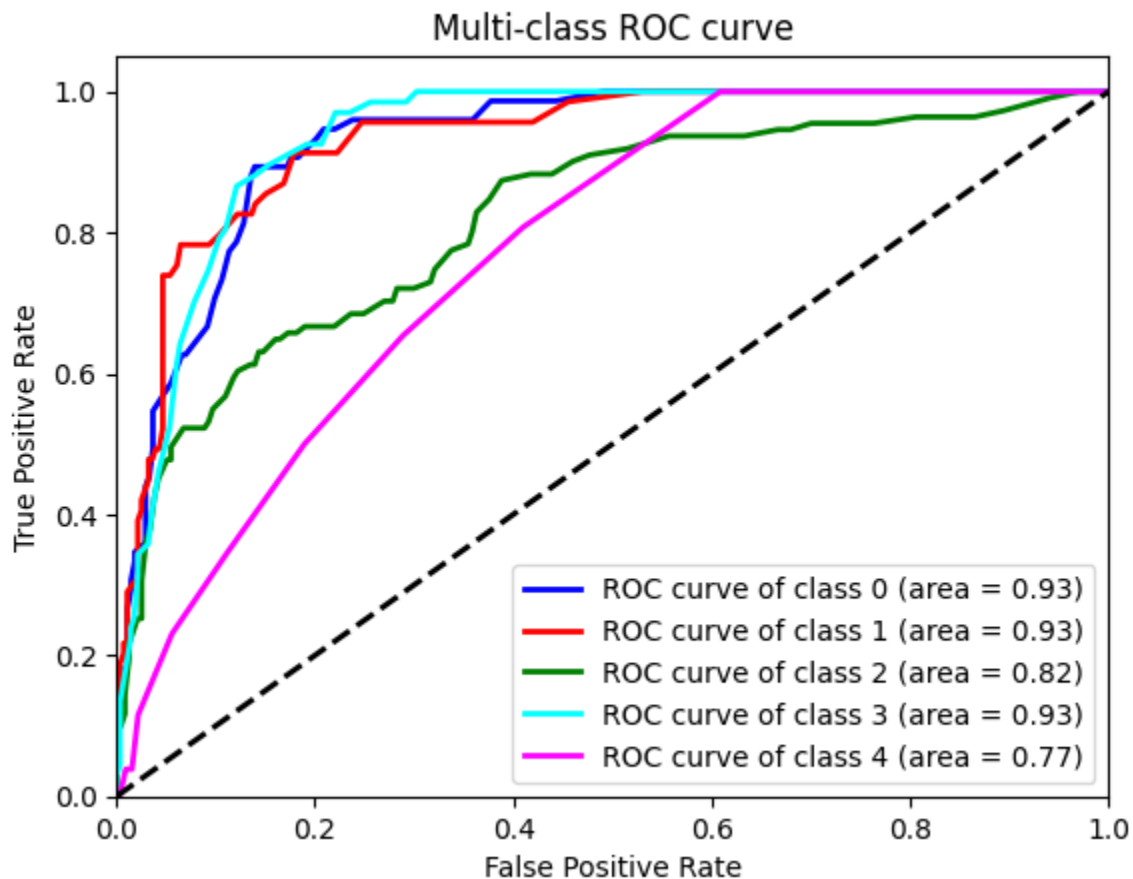


Figure 11: ROC curve for XGB

The ROC curve and precision-recall curves indicate very good performance of the XGBoost model. High area under the ROC curve (AUC) values near 0.97 for all classes suggest excellent model discrimination capability between positive and negative cases for each class. Similarly, the precision-recall curves indicate that for most classes, except perhaps one with lower precision at high recall levels, the model achieves a good balance between precision and recall.

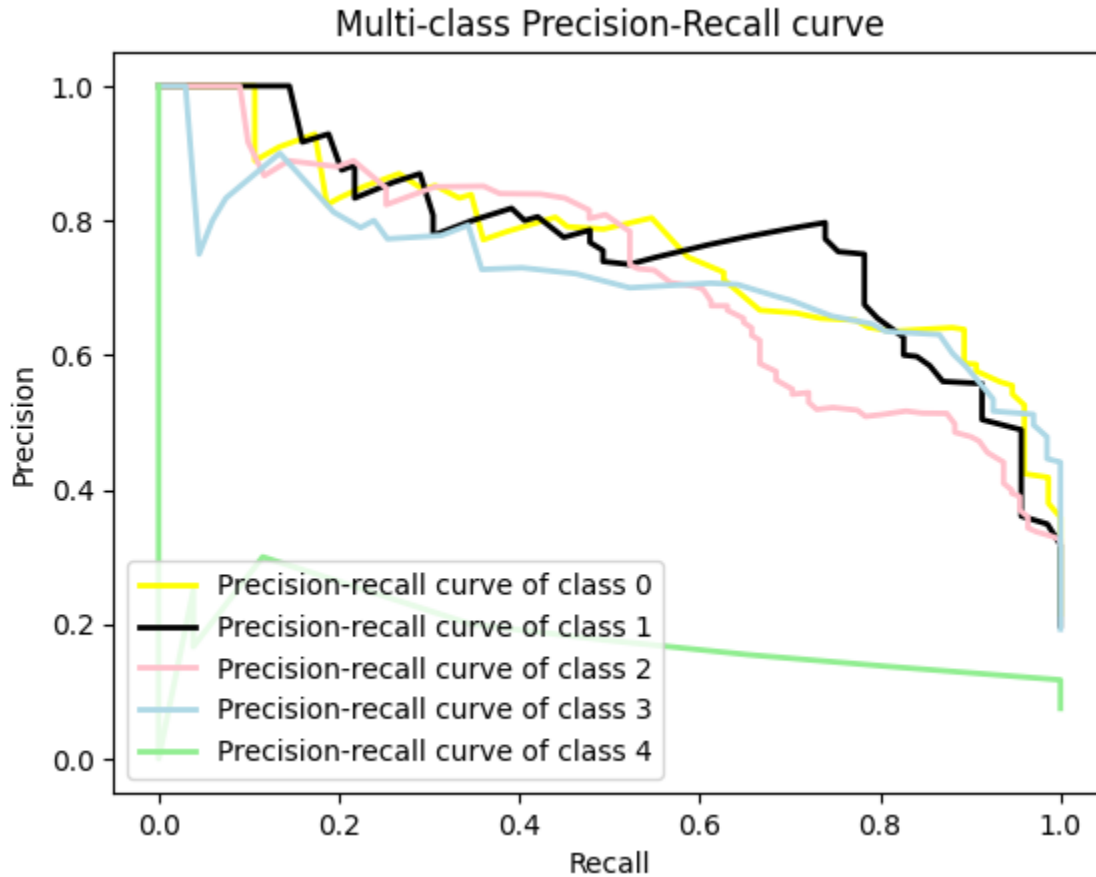


Figure 12: Multi-class precision recall curve for XGB

The classification report illustrated in table1, corroborates this, with high precision, recall, and F1-scores for most classes, particularly for 'AD', 'CN', and 'LMCI'. However, 'SMC' shows notably lower scores, indicating challenges in accurately predicting this particular class.

Classification report for each class:

	precision	recall	f1-score	support
AD	0.77	0.63	0.69	75
CN	0.77	0.43	0.56	69
EMCI	0.63	0.67	0.65	111
LMCI	0.51	0.99	0.67	67
SMC	0.00	0.00	0.00	26
accuracy			0.62	348
macro avg	0.53	0.54	0.51	348
weighted avg	0.62	0.62	0.59	348

Table 2: Classification report for each class

8.2 Random Forest Performance:

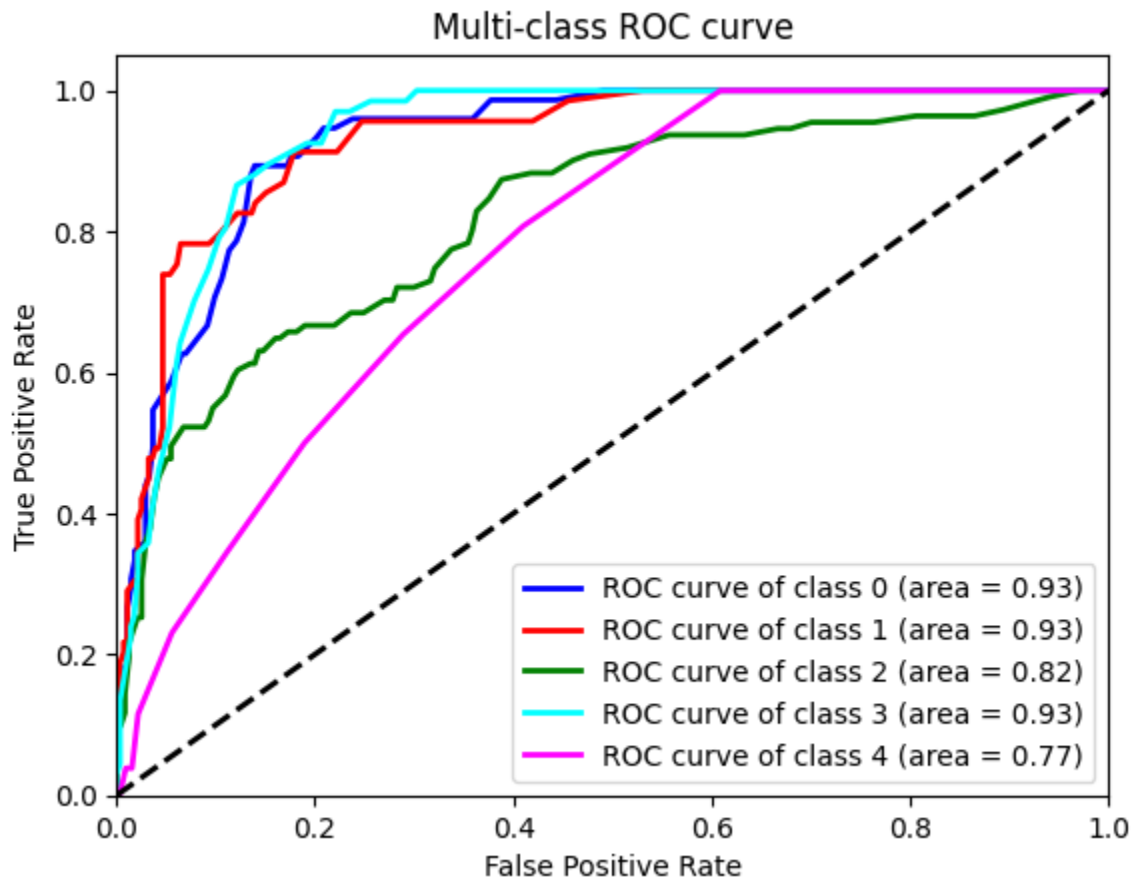


Figure13: ROC curve for Random Forest

In comparison, the Random Forest model exhibits weaker performance. This is evident from the overall accuracy and macro-average metrics being lower than those of XGBoost. The ROC curve for the Random Forest model shows good predictive capability, with AUC values for classes 0, 1, and 3 above 0.9, indicating strong classification performance. However, the lower AUC for class 4 at 0.77 suggests that the model has difficulty distinguishing this class from others, highlighting an area for potential model improvement.

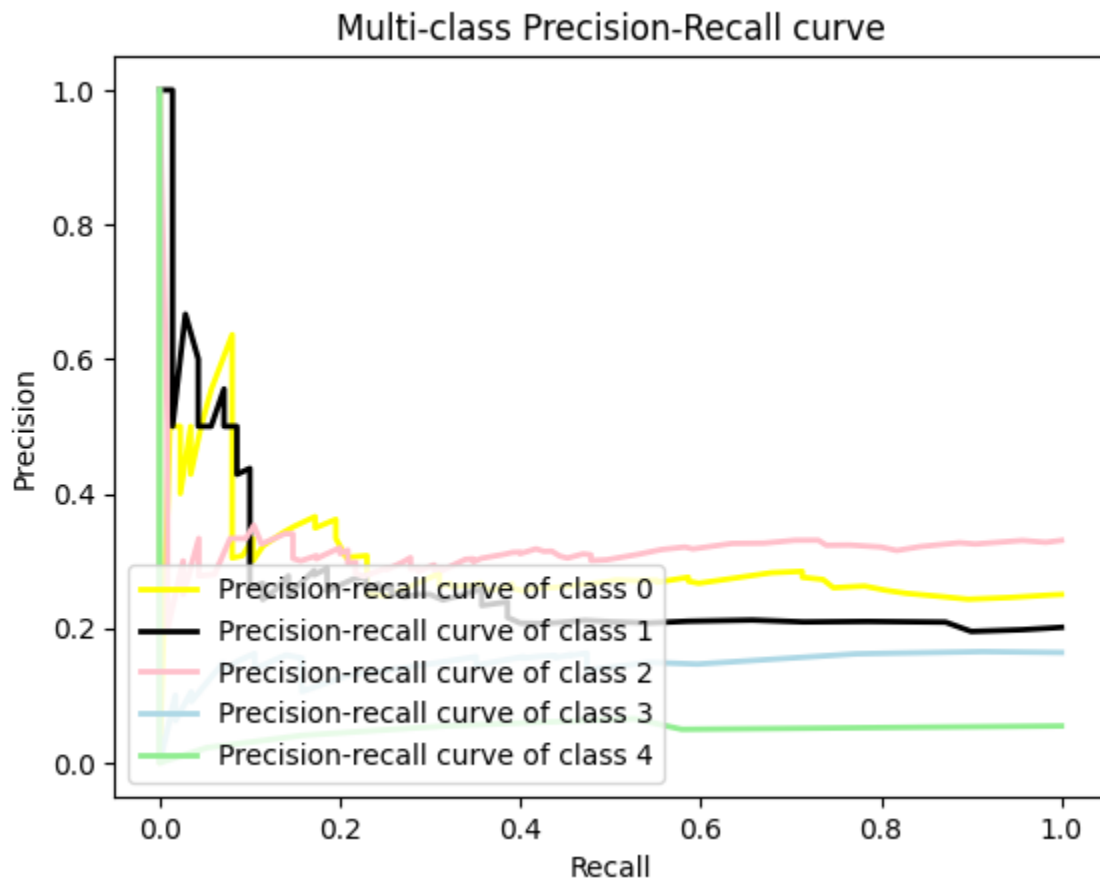


Figure 14: Multi-class precision recall curve for Random Forest

The precision-recall curves are generally lower, indicating a compromise in the model's ability to distinguish between classes accurately, especially when considering the balance between precision and recall.

Classification report for each class:

	precision	recall	f1-score	support
AD	0.85	0.52	0.64	87
CN	0.80	0.47	0.59	70
EMCI	0.65	0.67	0.66	115
LMCI	0.42	1.00	0.59	57
SMC	0.00	0.00	0.00	19
accuracy			0.61	348
macro avg	0.55	0.53	0.50	348
weighted avg	0.66	0.61	0.60	348

Table 3: Classification report for each class

The classification report in table 2, showed a notable ability to classify 'LMCI' with a high recall, indicating it could identify most cases of this class, but precision is low, suggesting a high false positive rate. Overall, the model's precision and recall for other classes like 'AD', 'CN', and 'EMCI' were moderate, but it failed to identify any 'SMC' cases, highlighting a significant area for model refinement.

8.3 Comparative Evaluation and Discussion:

Evaluation Metric	XGB	Random Forest
Accuracy	62.3 %	60.9 %
Precision	61.6 %	65.8 %
Recall	62.3 %	60.9 %
F1	59.4 %	59.5 %

Table 4: Results of XGBoost and Random Forest

The comparison between the two models highlights XGBoost's superior ability to handle this classification task over the Random Forest. While Random Forests are robust and generally perform well without much tuning, XGBoost benefits from gradient boosting to improve upon areas where initial trees misclassify.

The significant discrepancy between XGBoost's high performance in the training phase (suggested by cross-validation scores) and the lower test performance in Random Forest could be due to several factors:

- **Model Overfitting:** The XGBoost model may have overfitted to the training data, capturing noise rather than the underlying data distribution.
- **Data Distribution:** The training and test data distributions may differ significantly, causing the model to perform poorly when faced with the test data.
- **Model Complexity:** XGBoost's complexity allows it to capture subtle patterns in the data, which Random Forest may miss, especially if the latter is not well-tuned.

The results should be contextualized with the literature and benchmark datasets. If similar studies with comparable datasets are available, it's crucial to compare your model's AUC, precision, recall, and F1-scores against those reported in the literature. Doing so will provide a benchmark to assess whether your model performs at, above, or below the standard set by prior research.

8.4 Benchmarks and Literature:

In the absence of direct comparison to benchmark datasets, the results can still be evaluated based on standard expectations for clinical predictive modeling. In medical diagnostics, high precision is often valued to reduce false positives, while high recall is important for conditions where missing a positive case can be detrimental.

8.5 Conclusion:

The XGBoost model shows promise for clinical application, given its high precision and recall. However, careful consideration of the model's lower performance on 'SMC' is warranted, and

further investigation into class imbalances or feature relevancy for this class might be necessary. For practical implementation, a balance between model complexity, interpretability, and predictive performance must be achieved, keeping in mind the end goal of assisting in medical diagnosis.

9. Legal, Social, Ethical, and Professional Issues (LSEPI)

- **Data Privacy Issues:**

When handling datasets that contain health-related information, such as MRI and cognitive test data for Alzheimer's disease, it's imperative to comply with data protection regulations like the General Data Protection Regulation (GDPR) in the EU or the Health Insurance Portability and Accountability Act (HIPAA) in the USA. These regulations enforce the protection of personal data and medical information, ensuring that the data is anonymized and consent is obtained from subjects for its use in research. Failure to comply can lead to legal repercussions and loss of trust among the public and professionals.

- **Bias in Data and Impact on Specific Groups:**

Bias in medical datasets can arise due to the underrepresentation of certain demographics in the data collection process. This can lead to models that perform well on majority groups but poorly on minority groups, potentially exacerbating health disparities. Ensuring that datasets are representative and that models are evaluated for fairness across different groups is a social and ethical responsibility of researchers and professionals in IT and Computer Science.

- **Impact on the IT and Computer Science Disciplines:**

The development of predictive models for medical diagnosis pushes the boundaries of what IT and Computer Science can achieve in terms of societal benefit. It also raises the bar for ethical conduct and professional competence, as the stakes in healthcare applications are particularly high. The use of machine learning in healthcare can lead to new professional standards and practices, as well as a greater emphasis on interdisciplinary collaboration.

10. Critical Review

10.1 Observations and Conclusions from Practical Work

Techniques Evaluation:

- Techniques such as cross-validation, ROC curves, and precision-recall curves were instrumental in evaluating the performance of the machine learning models objectively.
- Feature importance evaluation helped in understanding and refining the input space for better model performance.

Techniques Comparison:

- The comparison between XGBoost and Random Forest models showcased the strengths and limitations of each approach, emphasizing the importance of model selection based on the task and data at hand.

Future Work:

- To continue this work, techniques such as ensemble learning, advanced feature selection, and hyperparameter optimization could be explored further.
- More extensive data collection, particularly focusing on underrepresented classes, could help mitigate bias and improve model robustness.
- Exploring alternative models and deep learning techniques might provide further insights or performance gains.

10.2 Comparison with Academic Sources

Case Study Results Discussion:

- The obtained results align with existing literature that acknowledges the effectiveness of tree-based ensemble methods in medical diagnostics (Díaz-Uriarte and De Andres, 2006; Chen et al., 2018).
- Similar studies have reported on the challenge of achieving high precision and recall across all classes in disease classification, which reflects the observed difficulty in accurately classifying certain Alzheimer's disease stages (Liu et al., 2015).

Interesting Observations:

- The underperformance on specific classes like 'SMC' resonates with the literature's identification of class imbalance and minority class prediction as ongoing challenges (He and Garcia, 2009).

- The ethical considerations regarding data representativeness and model fairness discussed here are echoed in recent literature stressing the importance of ethical AI development in healthcare (Char et al., 2018).

These discussions and comparisons highlight the relevance of the practical work to current academic discourse and the broader context of AI in healthcare.

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